Mohan Zhang

Ph.D. Student, College of AI, Tsinghua University
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RESEARCH INTERESTS

Reinforcement learning, large language models; currently focused on LLM agents, also interested in embodied AI.

EDUCATION

Tsinghua University, College of AI | Ph.D. in Artificial Intelligence — Sep 2026 – Present
Admitted through national recommendation for postgraduate study (entrance-exam exempt)

Beihang University | B.Eng., Image Processing Center — Sep 2022 – Jun 2026
GPA: **3.85/4.00** | Core courses: AI, Machine Learning, Digital Image Processing, Pattern Recognition, Computational Intelligence

PUBLICATIONS

- **M. Zhang**, C. You, X. Cao, Z. Sun, X. Jia, J. Zhou, Y. Chen. **R^2A: Learning Persona Policies Through Persona Representation Learning and Runtime Alignment.** **Submitted to AAAI 2027 (First Author)**. arXiv:2608.29798.
  Formalized persona behavior as a latent-state *selection–realization* process, exposing the selection and realization gaps of static persona prompting; proposed a two-stage framework (Who–How–What representation learning + feedback-driven runtime alignment) that outperforms the base model and static elicitation across 12 settings.
- N. Gao, X. Zhang, X. Jiang, M. You, **M. Zhang**, Y. Deng. **RF-Agent: Automated Reward Function Design via Language Agent Tree Search.** **NeurIPS 2025 (Spotlight)**.
  Developed an LLM agent framework that couples reward-code generation with tree search to automatically design and refine reward functions for reinforcement learning.
- X. Zhang, N. Gao, X. Jiang, Y. Chen, Y. Pan, **M. Zhang**, Y. Deng. **Progress Reward Model for Reinforcement Learning via Large Language Models.** **NeurIPS 2025 (Poster)**.
  Leveraged LLM world knowledge to decompose long-horizon tasks and generate subtask-progress functions, yielding a progress reward model that guides efficient policy learning.
- **M. Zhang**, J. Gao, S. Xu, Y. Wu. **Reasoning Is Not a Race: When Stopping Early Beats Going Deeper.** **NeurIPS 2025 (Poster, First Author)**.
  Provided a theoretical account of process collapse in PRM-guided beam search for long chain-of-thought reasoning, and designed an early-stopping rule based on locally normalized process-reward scores that improves accuracy while reducing compute.

RESEARCH EXPERIENCE

Apex Intelligence | Model Training Team, Intern — Jun 2026 – Nov 2026

- Deeply involved in training **Apex-35B-V1.0**, the company's first open-source large language model.

Ant Group | AReaL Team, Research Intern — 2025 – 2026

- Conducted research on reinforcement learning for large language models.

HONORS & AWARDS

Friends of Tsinghua – Huanlan Dream Scholarship, College of AI, Tsinghua University — Sep 2026
Baosteel Outstanding Student Award, Baosteel Education Foundation (2 undergraduate recipients at Beihang) — 2025
National Scholarship (twice), Ministry of Education of China (ranked 7/424 in 2023; 4/163 in 2024) — 2023, 2024
Outstanding Student of Beihang University (top 5%, three consecutive years) — 2023 – 2025
Competitions: Honorable Mention, MCM/ICM; First Prize, National College Students' Mathematics Competition; First Prize, National "Challenge Cup" Youth AI Competition; 10+ competition awards in total
Other: Merit Student, Outstanding Student Leader, Outstanding Graduate, etc.

SKILLS

Programming: Python, PyTorch | **Writing & Presentation:** LaTeX academic writing, scientific figures and slides
Languages: Chinese (native); English (CET-6), fluent in academic reading, paper writing, and conference presentation